

GOAL
Track how the **proportion** and **activation** of **reward-encoding neurons** change across **training days**, by **brain region** and **cohort** (supervised vs unsupervised).
Prediction: both rise in **anterior HVAs**, supervised only.

SHARED INPUTS

Neural data — up to ~80k neurons, SVD(400-PC)-compressed; reconstruct a large sample **per region** (V1 · medial · lateral · anterior)

Behavior — sound-cue position (≈ reward zone, present with OR without water), trial structure, corridor position, running

PRIMARY — d'(late vs early cue), across days

SAMPLE NEURONS per region per mouse (750 per region; capped at what a region has)

REWARD-ZONE ACTIVITY per rewarded-corridor (leaf1) trial, mean activity in the reward zone (5–40 dm)

SPLIT TRIALS by cue position
early-cue vs late-cue (cue tracks reward; exists in **both cohorts**, so the comparison is fair)

CROSS-VALIDATED d' (per neuron)
two interleaved trial folds;
 $d' = 2(\mu_{\text{late}} - \mu_{\text{early}})/(\sigma_{\text{late}} + \sigma_{\text{early}})$

CLASSIFY (flag)
reward-encoding if $|d'| \geq 0.3$ in **both** folds, same sign — noise rarely survives both

ACTIVATION (strength)
mean $|d'|$ over folds, for **all** neurons (unbiased)

AGGREGATE — mouse is the replicate
per mouse: % **reward-encoding** (how many) and **mean d'** (how strongly); mean ± s.e.m. across mice, across **days × region × cohort**

RIGOR / CONTROLS

cross-validation: effect must reproduce in both folds ·
shuffle null on late/early labels ·
cue-based reward event (cross-cohort fair) ·
mouse as replicate (neurons are not independent) ·
skip + report sessions with no cue variation or too few trials

SHUFFLE FLOOR
scramble late/early labels, rerun the *same* cross-validated test → chance % expected from noise

INTERPRETATION
Above the chance floor, rising before → after learning, concentrated in **anterior HVAs**, **supervised only** ⇒ supports the reward-prediction account. Unsupervised should stay near the floor.